Q3		Suggested Solutions	Mark
3	(a)	The incident wave from the oscillator is **reflected at Q.** The incident and reflected waves have the **same speed, wavelength (or frequency) and amplitude, travelling in opposite directions overlap/meet/superpose** and form a stationary wave.	B1 B1
	(b)	Q is a **fixed end or node**. **Destructive interference** must occur, so that the **net displacement is always zero**. Hence there must be a phase change of 180° (or π radians) at Q.	B1
	(c)	**(i)** $\lambda/2 = 0.30$ m $\lambda = 0.60$ m $v = f\lambda = (120)(0.60)$ $= 72$ m s^{-1}	 M1 A1
		(ii) $v_0 = \omega x_0 = 2\pi(120)(0.0080)$ $= 6.03$ m s^{-1}	M1 A1

Qn 4	**Suggested Solutions**	**Mark**
4(a)	The electric potential at a point is the **work done per unit positive charge**	**B1**